# Coverage Maximization for Heterogeneous Aerial Networks

Daosen Zhai, Qiqi Shi, Ruonan Zhang, *Member, IEEE*, Xiao Tang, *Member, IEEE*, and Haotong Cao

***Abstract*—In this letter, we investigate the network planning of the heterogeneous aerial networks to maximize the network coverage. Specifically, a joint high-altitude aerial base station (HABS) trajectory and low-altitude aerial base station (LABS) position optimization problem is formulated with the objective to maximize the number of covered users. Based on the feature of the air-to-ground channel, we first obtain the optimal altitude of the LABSs through the bisection method. Then, we propose a joint particle-and-fish swarm algorithm to optimize the horizontal position of the LABS. For the uncovered users, we model the user coverage probability in the HABS and devise a random interval contraction based algorithm to optimize the flying radius of the HABS. Simulation results indicate that our algorithms can fully exploit the mobility of the heterogeneous aerial networks to cover more users.**

***Index Terms*—Aerial networks, unmanned aerial vehicle, network coverage, optimization algorithm.**

## I. Introduction

DUE TO the flexibility, the unmanned aerial vehicle (UAV) based wireless networks have great application potential in the field of military and civilian. As an important application, UAV equipped with communication devices can be used as aerial base station (ABS) to provide wide-area coverage for remote region. In comparison with the terrestrial networks, the light-of-sight (LoS) channel can be more likely established in the air-to-ground communication [1], as less obstacles exist between the ABS and the ground users. Furthermore, the UAV can be dynamically deployed to match the user distribution and thereby improve the network performance, which is incomparable to the terrestrial networks. These advantages promote the application of UAVs in the future wireless networks, and it has been predicted that the aerial network will be an important component of the sixth generation mobile communication systems (6G) [2].

How to deploy the ABSs (e.g., position and trajectory) is vital for the network performance, which has inspired abundant researches. In [3], the authors proposed an on-demand density-driven UAV three-dimensional (3D) placement algorithm to cover as much users as possible subject to the constraint of the minimum rate requirement. In [4], the authors proposed a flow-level model (FLM) to realistically characterize the ABS performance, and adopted the deep reinforcement learning (DRL) approach to optimize the UAV trajectory. In [5], UAVs were deployed to help the terrestrial base stations (BSs) to improve the user coverage probability. In [6], a dual-band airborne access network was proposed. The channel characteristics in the dual band were measured, based on which the authors devised a joint channel-and-power allocation algorithm. Some ABSs related prototyping and field tests (i.e., Facebook Aquila, Google Loon, Nokia F-Cell, etc.) were introduced in [7]. Multiple access techniques especially the non-orthogonal multiple access (NOMA) are important in UAV-enabled wireless networks [8], [9]. In [10], a new NOMA based UAV relay protocol was proposed to enhance the coverage for the cell-edge users in the terrestrial cellular networks. In addition, the UAVs can also be utilized as data collector for sensors [11], [12].

Although abundant optimization algorithms have been proposed for the ABSs, few works focus on the heterogeneous aerial networks (HetANs). In practice, the high-altitude ABS (HABS) with low frequency can achieve wide-area coverage, while the low-altitude ABS (LABS) with high frequency can support high-capacity communications. Therefore, the HABS cooperates with the LABS can provide better service in comparison with the homogeneous aerial networks (HomANs). However, the coverage problem in the HetANs is more complicated than that in the HomANs [3], [5], as the joint deployment optimization is needed which has not been well studied.

Motivated by the above, we investigate the coverage maximization problem via network planing for the HetANs. The contributions of the letter is summarized as follow.

- We propose a HetAN, where a HABS installed on a fixed-wing UAV cooperates with multiple LABSs installed on some rotary-wing UAVs to provide service for the ground users in a large region.
- Based on the bionics and optimization algorithms, we propose two efficient algorithms to optimize the LABS-position and the HABS-trajectory to maximize the number of covered users.
- We conduct simulations to evaluate the effect of the 3D-position of the LABS and the trajectory of the HABS on the network coverage, and demonstrate the efficiency of the proposed algorithms.

Manuscript received September 3, 2021; accepted October 14, 2021. Date of publication October 19, 2021; date of current version January 10, 2022. This work was supported in part by the National Key Research and Development Program of China under Grant 2020YFB1807003; in part by the National Natural Science Foundation of China under Grant 61901381, Grant 61901378, and Grant 62171385; in part by the Aeronautical Science Foundation of China under Grant 2020Z073053004; in part by the Key Research Program and Industrial Innovation Chain Project of Shaanxi Province under Grant 2019ZDLGY07-10; and in part by the Foundation of the State Key Laboratory of Integrated Services Networks of Xidian University under Grant ISN21-06. The associate editor coordinating the review of this article and approving it for publication was A. Shojaeifard. *(Corresponding author: Ruonan Zhang.)*

Daosen Zhai is with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an 710072, Shaanxi, China, and also with the State Key Laboratory of Integrated Services Networks, Xidian University, Xi'an 710071, China (e-mail: zhaidaosen@nwpu.edu.cn).

Qiqi Shi, Ruonan Zhang, and Xiao Tang are with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an 710072, Shaanxi, China (e-mail: shiqiqi@mail.nwpu.edu.cn; rzhang@nwpu.edu.cn; tangxiao@nwpu.edu.cn).

Haotong Cao is with the Department of Computing, Hong Kong Polytechnic University, Hong Kong, SAR, China (e-mail: haotong.cao@polyu.edu.hk).

Digital Object Identifier 10.1109/LWC.2021.3121076

2162-2345 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hanoi University of Science & Technology. Downloaded on March 12,2026 at 08:52:19 UTC from IEEE Xplore. Restrictions apply.

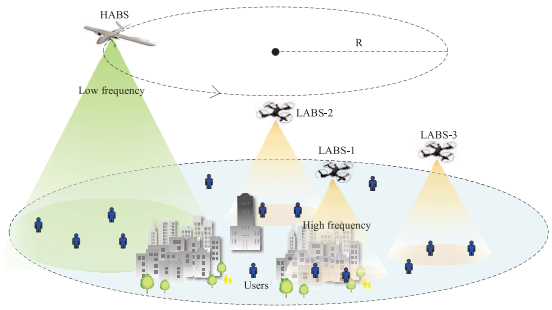


Fig. 1. A heterogeneous aerial network with a HABS and multiple LABSs.

## II. Network Model and Problem Formulation

In this section, we first introduce the HetAN model, and then give the problem formulation.

### A. Network Model

As depicted in Fig. 1, we consider a HetAN, which consists of a HABS, $N$ LABSs denoted by $\mathcal{N}$, and $M$ terrestrial users denoted by $\mathcal{M}$. The HABS periodically flies at a constant velocity and along a prescribed circumference with fixed altitude $H$ and a fixed center to cover a ground circular region with radius $R_S$, wherein the $M$ users are uniform distributed. Without loss of generality, the coordinate of the center is defined as $w_0 = (0, 0, H)$. The flying radius of the HABS is denoted by $R$, which can be adjusted in a certain range. For wide-area coverage, the low frequency (LF) band consisted of $K_l$ channels is adopted by the HABS. Besides, $N$ LABSs using high frequency (HF) band consisted of $K_h$ channels are dynamically deployed to further improve the network coverage. The channel bandwidth of the LF band is defined as $B_l$, and that of the HF band is defined as $B_h$. The position of LABS $n$ is denoted by $w_n = (x_n, y_n, h_n)$, and the set of all the position of LABSs is represented by $\mathbf{W} = \{w_n\}$. The position of user $m$ is denoted by $w_m = (x_m, y_m, h_m)$. The users access to the ABS with the maximum receive signal strength indicator (RSSI).

In this letter, we only consider the path loss, which can be seen as the time-averaged performance. According to [13], the air-to-ground path loss consists of LoS and non-LoS (NLoS) components. The LoS probability between ABS $n$[1] and user $m$ is given by

$$P_{LoS}(n,m) = \frac{1}{1 + \alpha e^{\left(-\beta\left(\arctan\left(\frac{h_n - h_m}{d_{n,m}^{2D}}\right) - \alpha\right)\right)}}, \quad (1)$$

where $\alpha$ and $\beta$ are environmental parameters, and $d_{n,m}^{2D} = \sqrt{(x_n - x_m)^2 + (y_n - y_m)^2}$.

For a typical urban macro (UMa) [14], the LoS and NLoS models are respectively given by

$$PL_{LoS} = 32.44 + 20\lg(d[m]) + 20\lg(f_c[GHz]), \quad (2)$$
$$PL_{NLoS} = 32.4 + 30\lg(d[m]) + 20\lg(f_c[GHz]), \quad (3)$$

where $d$ denotes the distance, and $f_c$ denotes the frequency.

[1]We use $n = 0$ to represent the HABS, and $n > 0$ to represent the LABS.

According to the above, the path loss from ABS $n$ to user $m$ can be calculated by

$$PL_{n,m} = P_{LoS} PL_{LoS} + (1 - P_{LoS}) PL_{NLoS}. \quad (4)$$

The received power of user $m$ from ABS $n$ is given by

$$Pr_m = Pt_n - PL_{n,m}, \quad (5)$$

where $Pt_n$ denotes the transmission power of ABS $n$.

To evaluate the network coverage performance, we define the coverage probability of user $m$ as

$$\gamma_m = P(Pr_m \geq \Gamma), \quad (6)$$

where $\Gamma$ denotes the threshold power that depends on the receiver sensitivity.

### B. Problem Formulation

The coverage probability $\gamma_m$ is related to the positions of the LABSs $\mathbf{W}$ and the trajectory of the HABS. Since the HABS can only fly along a circumference with a fixed center, the trajectory optimization of the HABS becomes to find the optimal flying radius $R$. Thus, network planning by jointly optimizing $\mathbf{W}$ and $R$ is needed for coverage maximization. In light of this, we formulate the following problem.

$$\begin{aligned} \max_{\mathbf{W},R} \quad & \sum_{m \in \mathcal{M}} \gamma_m \\ \text{s.t.} \quad & \text{C1}: x_{\min} \leq x_n \leq x_{\max}, \forall n \in \mathcal{N} \\ & \text{C2}: y_{\min} \leq y_n \leq y_{\max}, \forall n \in \mathcal{N} \\ & \text{C3}: h_{\min} \leq h_n \leq h_{\max}, \forall n \in \mathcal{N} \\ & \text{C4}: R_{\min} \leq R \leq R_{\max}, \end{aligned} \quad (7)$$

where $x_{\min}$, $x_{\max}$, $y_{\min}$, $y_{\max}$, $h_{\min}$, $h_{\max}$, $R_{\min}$ and $R_{\max}$ are the boundary values of the corresponding variables.

In (7), the objective function is to maximize the total coverage probability of all users. Constraints C1-C3 limit the 3D deployment area for each LABS. Constraint C4 specifies the value range of the flying radius of the HABS.

## III. Algorithm Design

To tackle the problem in (7), we divide it into two sequential subproblems (i.e., the LABS position optimization and the HABS flying radius optimization) and then devise two efficient algorithms to solve them respectively.

### A. LABS Position Optimization

In this section, we focus on the LABS position optimization problem. According to the network model, we know that given the threshold power $\Gamma$, there exists a maximum coverage area for each LABS. When the user is located in the coverage area, its coverage probability is equal to one, otherwise it is equal to zero. As such, the LABS position optimization problem becomes the max-vertex-covering problem.

The channel model in (4) indicates that there is an optimal altitude for the LABS to achieve the maximum coverage radius $r_{LABS}^{\max}$. It has been discussed that the coverage radius is an

Authorized licensed use limited to: Hanoi University of Science & Technology. Downloaded on March 12,2026 at 08:52:19 UTC from IEEE Xplore. Restrictions apply.

**Algorithm 1** The Joint Particle-and-Fish Swarm Algorithm

1: Initialize $\omega = 0.9$, $C_1 = C_2 = 2$, and the maximum number of iterations $J = 200$.
2: Randomly generate a set of positions and velocities $(X_1, V_1)$ of the particle swarm and the positions $X_2$ of the fish swarm in the feasible region.
3: Calculate the number of covered users $Y_1$ at the positions $X_1$, and denote the max-value as *t*; Calculate the number of covered users $Y_2$ at the positions $X_2$, and denote the max-value as *s*.
4: Compare $Y_1$ with $Y_2$ termwise, and write the better position $P^{\text{best}}$ into the bulletin board.
5: Compare *t* with *s*, and write the best position $G^{\text{best}}$ into the bulletin board.
6: **for** $j = 1 : J$ **do**
7: Calculate the particle velocity $V_1^{\text{new}}$ and position $X_1^{\text{new}}$ according to (9) and (10).
8: Calculate the number of covered users $Y_1^{\text{new}}$ at the position $X_1^{\text{new}}$.
9: Compare $Y_1^{\text{new}}$ with the data in the bulletin board termwise, and update $P^{\text{best}}$ if the former is bigger.
10: Compare the max-value in $Y_1^{\text{new}}$ with the data in the bulletin board, and update $G^{\text{best}}$ if the former is bigger.
11: Find the next position of the fish swarm $\left(X_{2,1}, X_{2,2}\right)$ by the AF-swarm behavior and the AF-follow behavior respectively [**?**], and calculate the corresponding number of covered users $\left(Y_{2,1}^{\text{new}}, Y_{2,2}^{\text{new}}\right)$.
12: Compare the max-value in $\left(Y_{2,1}^{\text{new}}, Y_{2,2}^{\text{new}}\right)$ with that in the bulletin board, and update $G^{\text{best}}$ if the former is bigger.
13: **end for**
14: Obtain the positions of the LABSs as $G^{\text{best}}$.

unimodal function in the altitude [15]. Thus, we can adopt the bisection method to quickly obtain the optimal altitude $h_n^*$ for each LABS. After getting $h_n^*$ and $r_{LABS}^{\max}$, we can optimize the horizontal position of the LABS to maximize the number of covered users. The position optimization problem of the LABSs can be recast as

$$\begin{aligned} \max_{\{(\mathrm{x_n},\mathrm{y_n})\}} \quad & \sum_{m\in\mathcal{M}} I_m\left(\min_{n\in\mathcal{N}} d_{n,m}^{2D} \le r_{LABS}^{\max}\right) \\ \text{s.t.} \quad & \text{C1}: x_{\min} \le x_n \le x_{\max}, \forall n \in \mathcal{N} \\ & \text{C2}: y_{\min} \le y_n \le y_{\max}, \forall n \in \mathcal{N}, \end{aligned} \tag{8}$$

where $I_m(\min_{n\in\mathcal{N}} d_{n,m}^{2D} \le r_{LABS}^{\max})$ is an indicator function that if the condition is satisfied, $I_m = 1$, otherwise $I_m = 0$.

The LABSs can be deployed in a continuous position, which results in a very complicated solving process. So, we adopt the bionics algorithm, which is especially suitable for solving the large-scale complex optimization problems. Specifically, we combine the particle swarm optimization (PSO) and the fish swarm optimization (FSO). The PSO has fast convergence speed, but it is easy to fall into local optimum. The FSO has good global convergence feature, but the convergence speed is slow. We combine their advantages to design an effective algorithm, which is summarized in Algorithm 1.

The update formulas of the velocity and the position in the PSO are given by

$$\begin{aligned} V_i(t+1) = \omega V_i(t) + C_1 \text{random}\ (0,1)\Big(P_i^{best}(t) - X_i(t)\Big) \\ + C_2 \text{random}\ (0,1)\Big(G^{best}(t) - X_i(t)\Big), \end{aligned} \tag{9}$$

$$X_i(t+1) = X_i(t) + V_i(t+1), \tag{10}$$

where $\omega$ represents the weight of inertia, $V_i(t)$ represents the velocity of particle *i* at time *t*, $X_i(t)$ represents the position of particle *i* at time *t*, $P_i^{best}(t)$ is the best position experienced by particle *i*, $G^{best}(t)$ is the best position experienced by all the particles, $C_1$ and $C_2$ are the acceleration coefficients (generally set as 2). The first part of the velocity update formula (9) reflects the effect of the inertia of the previous velocity. The second part shows the influence of the particle's own information on its next behavior. The third part shows the information sharing and cooperation among particles, i.e., the influence of the swarm on the next behavior of particles.

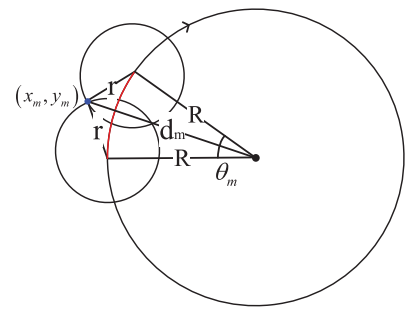


Fig. 2. An illustration for the coverage of the users in the HABS.

### B. HABS Flying Radius Optimization

For the uncovered users, we deploy the HABS to serve them as much as possible. Different from the LABSs, the position of the HABS is dynamic. As a consequence, the coverage for the users in the HABS is time-incontinuous, that is, the coverage probability $\gamma_m$ is smaller than one.

To quantify $\gamma_m$, we depict it in Fig. 2, where *r* denotes the coverage radius of the HABS (noted that *r* is a determined value when the system parameters are given), and $d_m = \sqrt{x_m^2 + y_m^2}$. If user *m* is located in the coverage of the HABS, its severing time corresponds to the red arc or the angle $\theta_m$ in Fig. 2. Using the geometrical relationship, we get

$$\theta_m = 2\arccos\left(\frac{R^2 + d_m^2 - r^2}{2Rd_m}\right). \tag{11}$$

The coverage probability of user *m* in the HABS is

$$\gamma_m(R) = \begin{cases} \frac{\theta_m}{2\pi}, & \text{if } R - r < d_m < R + r \\ 0, & \text{otherwise.} \end{cases} \tag{12}$$

Then, the coverage maximization problem of the HABS is transformed into finding the maximum value of the following function in $[R_{\min}, R_{\max}]$.

$$f(R) = \sum_{m\in\mathcal{M}\setminus\mathcal{M}_L} \gamma_m(R), \tag{13}$$

where $\mathcal{M}_L$ denotes the users covered by the LABSs. Since $f(R)$ is an unary continuous function, we propose the random interval contraction based algorithm, which is summarized in Algorithm 2. The main idea is that we first uniformly choose *K* points in the feasible region $[R_{\min}, R_{\max}]$ (step 2), and find the one $\hat{R}$ that leads to the maximum objective function value (step 3). Then, during the neighborhood of $\hat{R}$ (step 4), we adopt the golden section method to shrink the feasible region (steps 5-14) until satisfying the termination condition (step 6) and outputting the flying radius (step 15).

Authorized licensed use limited to: Hanoi University of Science & Technology. Downloaded on March 12,2026 at 08:52:19 UTC from IEEE Xplore. Restrictions apply.

**Algorithm 2** The Random Interval Contraction-Based Algorithm

Initialize the number of points $K = 100$, the error tolerance $\varepsilon = 0.1$, and the golden ratio $\mu = \frac{\sqrt{5}-1}{2}$.
Uniformly select $K$ points in $[R_{\min}, R_{\max}]$ and calculate the objective function values (OFV) $f(R_k)$.
Find the maximum value from the $K$ OFVs and denote the corresponding solution as $\hat{R}$.
Set the left margin $a = \max\left\{\hat{R} - \frac{\Delta}{K}, R_{\min}\right\}$ and the right margin $b = \min\left\{\hat{R} + \frac{\Delta}{K}, R_{\max}\right\}$, where $\Delta = R_{\max} - R_{\min}$.
Choose two points in [a, b] as $R_1 = a + (1-\mu)(b-a)$ and $R_2 = a + \mu(b-a)$, and calculate $f(R_1)$ and $f(R_2)$.
**while** $b - a > \varepsilon$ **do**
  **if** $f(R_1) - f(R_2) < 0$ **then**
    Set $a = R_1$, $R_1 = R_2$, and $R_2 = a + \mu(b-a)$.
    Calculate $f(R_2)$.
  **else**
    Set $b = R_2$, $R_2 = R_1$, and $R_1 = a + (1-\mu)(b-a)$.
    Calculate $f(R_1)$.
  **end if**
**end while**
Output the flying radius as $R^* = \frac{a+b}{2}$.

TABLE I
SIMULATION PARAMETERS

| | |
|---|---|
| Altitude of the HABS, $H$ | 1 km [7] |
| Altitude of the LABSs, $h_n$ | [50 m, 500 m] [7] |
| x-coordinate of the LABS, $x_n$ | [-$R_S$, $R_S$] |
| y-coordinate of the LABS, $y_n$ | [-$R_S$, $R_S$] |
| Flying radius, $[R_{\min}, R_{\max}]$ | $[r, R_S + r]$ |
| Low frequency | 785 MHz [6] |
| High frequency | 2160 MHz [6] |
| Transmission Power, $Pt_n$ | 5 Watt [17] |
| Channel bandwidth, $B_l$ | 180 KHz [6] |
| Channel bandwidth, $B_h$ | 900 KHz [6] |
| Channel numbers, $K_l$ and $K_h$ | 350 [6] |
| Number of the LABSs, $N$ | 3 |
| Environmental parameter, $\alpha$ | 9.61 [17] |
| Environmental parameter, $\beta$ | 0.16 [17] |
| Simulation times | 500 |

## IV. SIMULATION RESULTS

In this section, the simulation results are provided to evaluate the performance of our scheme (referred to as the Joint Optimization). In the simulation scenario, the positions of the users are randomly generated and follow the uniform distribution. The common simulation parameters are listed in Table I, and the others are given in the simulation. In order to demonstrate the advantages of our scheme, we compare it with the other four schemes, namely Random Deployment, w/o Altitude Optimization, w/o Horizontal Position Optimization, and w/o Flying Radius Optimization, the introduction of which is given as follows.

- *Random Deployment:* In this scheme, the 3D positions of the LABSs and the flying radius of the HABS are randomly generated in the feasible region.
- *w/o Altitude Optimization:* This scheme optimizes the horizontal positions of the LABSs and the flying radius of the HABS by our proposed algorithms. However, the altitude of the LABSs is randomly generated.
- *w/o Horizontal Position Optimization:* This scheme optimizes the altitude of the LABSs and the flying radius of the HABS, but the horizontal position of the LABSs is randomly generated.
- *w/o Flying Radius Optimization:* In this scheme, the flying radius of the HABS is randomly generated, and the other parameters are optimized by our algorithms.

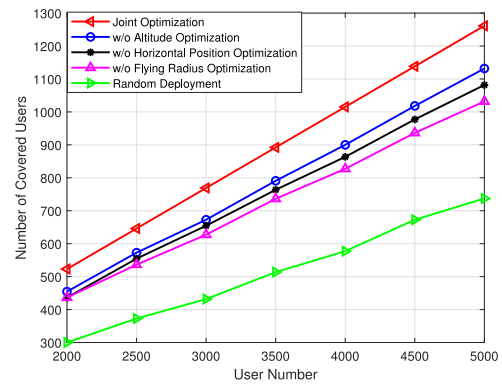


Fig. 3. The number of covered users versus the total user number $M$ ($\Gamma = -90$ dBm and $R_S = 3$ km).

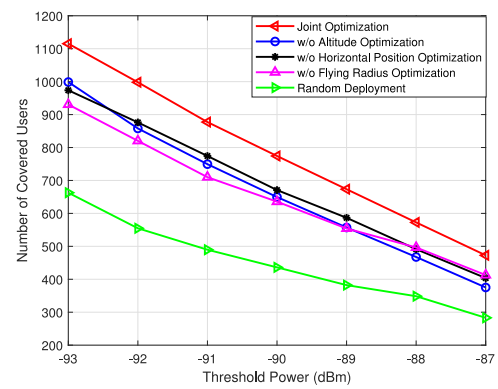


Fig. 4. The number of covered users versus the threshold power $\Gamma$ ($M = 3000$ and $R_S = 3$ km).

Fig. 3 shows the impact of the user number $M$ on the network coverage. From this figure, we can observe that our scheme can greatly enhance the network performance. Compared with the Random Deployment, the number of covered users through the joint position-and-trajectory optimization is upgraded by about 70%. This verify the efficiency of our proposed algorithms. Besides, the simulation results also confirm the significance of the ABSs' altitude, horizontal position, and flying radius, where the flying radius has the greatest impact. This is because partial users are covered by the LABSs, and hence the distribution of the remainder users becomes nonuniform. To cover more users, the HABS should avoid flying over the blank area as much as possible. Therefore, the flying radius of the HABS is very important for coverage maximization. This result also reflects the necessity of the cooperation between the HABS and the LABSs.

Fig. 4 depicts the number of covered users versus the threshold power $\Gamma$. It can be seen from the figure that the number of covered users decreases with $\Gamma$. This is because with the increment of $\Gamma$, the coverage radius of both the LABSs and the HABS decreases. When $\Gamma$ is small, the performance gain by optimizing the HABS trajectory is larger than that by

Authorized licensed use limited to: Hanoi University of Science & Technology. Downloaded on March 12,2026 at 08:52:19 UTC from IEEE Xplore. Restrictions apply.

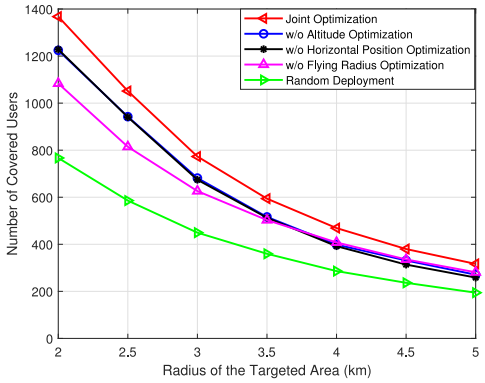


Fig. 5. The number of covered users versus the radius of the target area $R_S$ ($M = 3000$ and $\Gamma = -90$ dBm).

optimizing the LABS position. When $\Gamma$ is large, the situation is reverse. The reason is that when $\Gamma$ is large, few users are covered by the LABSs, and thus the distribution of the remainder users is almost uniform. Therefore, the effect by optimizing the trajectory of the HABS becomes small. In practice, the ABS is usually utilized for low-rate and wide-area coverage, that is $\Gamma$ is relatively low. Therefore, the joint position-and-trajectory optimization is necessary in practice.

Fig. 5 shows the effect of the radius of the targeted area $R_S$ on the number of covered users. Firstly, we can find that the number of covered users decreases with $R_S$. The reason is identical to that in Fig. 3. Besides, this figure shows that with any $R_S$, our scheme can achieve the best performance and greatly outperform the other schemes, which fully demonstrate the efficiency of our proposed algorithms. The simulation results indicate that the altitude and the horizontal position of the LABS are equally important under this simulation parameters, especially when the targeted area is large. Moreover, the flying radius of the HABS is still the most important factor for coverage maximization.

## V. Conclusion

In this letter, we have studied the coverage maximization problem for the heterogeneous aerial networks. Specifically, we have formulated a joint HABS-trajectory and LABS-position optimization problem to maximize the number of covered users. To solve this hard problem, we have first optimized the 3D position of the LABSs based on the channel feature and the particle-and-fish swarm optimization. Then, we have devised a random interval contraction based algorithm to optimize the trajectory of the HABS. Finally, we have conducted simulations to verify the importance of the control policies and the efficiency of our algorithms.

In this letter, we assume that the positions of the users are fixed. How to cover the mobile users is a valuable research direction. Another interesting direction is considering the random trajectory for the HABS. Furthermore, the backhaul link should be carefully design for practical application.

## References

[1] Y. Zeng, R. Zhang, and T. J. Lim, "Wireless communications with unmanned aerial vehicles: Opportunities and challenges," *IEEE Commun. Mag.*, vol. 54, no. 5, pp. 36–42, May 2016.

[2] X. You *et al.*, "Towards 6G wireless communication networks: Vision, enabling technologies, and new paradigm shifts," *Sci. China Inf. Sci.*, vol. 64, no. 1, Jan. 2021, Art. no. 110301.

[3] C.-C. Lai, C.-T. Chen, and L.-C. Wang, "On-demand density-aware UAV base station 3D placement for arbitrarily distributed users with guaranteed data rates," *IEEE Wireless Commun. Lett.*, vol. 8, no. 3, pp. 913–916, Jun. 2019.

[4] V. Saxena, J. Jaldn, and H. Klessig, "Optimal UAV base station trajectories using flow-level models for reinforcement learning," *IEEE Trans. Cogn. Commun. Netw.*, vol. 5, no. 4, pp. 1101–1112, Dec. 2019.

[5] Z. Chen and H. Zhang, "UAV-assisted networks through a tunable dependent model," *IEEE Commun. Lett.*, vol. 24, no. 5, pp. 1110–1114, May 2020.

[6] R. Zhang, Q. Guo, D. Zhai, D. Zhou, X. Du, and M. Guizani, "Channel measurement and resource allocation scheme for dual-band airborne access networks," *IEEE Access*, vol. 7, pp. 80870–80883, 2019.

[7] A. Fotouhi *et al.*, "Survey on UAV cellular communications: Practical aspects, standardization advancements, regulation, and security challenges," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 4, pp. 3417–3442, 4th Quart., 2019.

[8] Y. Cao *et al.*, "Secure transmission via beamforming optimization for NOMA networks," *IEEE Wireless Commun.*, vol. 27, no. 1, pp. 193–199, Feb. 2020.

[9] W. Bao, H. Chen, Y. Li, and B. Vucetic, "Joint rate control and power allocation for non-orthogonal multiple access systems," *IEEE J. Sel. Areas Commun.*, vol. 35, no. 12, pp. 2798–2811, Dec. 2017.

[10] D. Zhai, H. Li, X. Tang, R. Zhang, Z. Ding, and F. R. Yu, "Height optimization and resource allocation for NOMA enhanced UAV-aided relay networks," *IEEE Trans. Commun.*, vol. 69, no. 2, pp. 962–975, Feb. 2021.

[11] M. Yi, X. Wang, J. Liu, Y. Zhang, and B. Bai, "Deep reinforcement learning for fresh data collection in UAV-assisted IoT networks," in *Proc. IEEE INFOCOM Conf. Comput. Commun. Workshops (INFOCOM WKSHPS)*, 2020, pp. 716–721.

[12] W. Chen, S. Zhao, R. Zhang, Y. Chen, and L. Yang, "UAV-assisted data collection with nonorthogonal multiple access," *IEEE Internet Things J.*, vol. 8, no. 1, pp. 501–511, Jan. 2021.

[13] A. Al-Hourani, S. Kandeepan, and A. Jamalipour, "Modeling air-to-ground path loss for low altitude platforms in urban environments," in *Proc. IEEE Global Commun. Conf.*, 2014, pp. 2898–2904.

[14] "Study on channel model for frequencies from 0.5 to 100 GHz (release 14), v14.0.0," 3GPP, Sophia Antipolis, France, Rep. TR 38.901, Jul. 2017.

[15] A. Al-Hourani, S. Kandeepan, and S. Lardner, "Optimal LAP altitude for maximum coverage," *IEEE Wireless Commun. Lett.*, vol. 3, no. 6, pp. 569–572, Dec. 2014.

[16] Y. Feng, S. Zhao, and H. Liu, "Analysis of network coverage optimization based on feedback k-means clustering and artificial fish swarm algorithm," *IEEE Access*, vol. 8, pp. 42864–42876, 2020.

[17] Y. Song, S. H. Lim, S.-W. Jeon, and S. Baek, "On cooperative achievable rates of UAV assisted cellular networks," *IEEE Trans. Veh. Technol.*, vol. 69, no. 9, pp. 9882–9895, Sep. 2020.

Authorized licensed use limited to: Hanoi University of Science & Technology. Downloaded on March 12,2026 at 08:52:19 UTC from IEEE Xplore. Restrictions apply.